MIL-D-51388A(EA)
14 November 1979
SUPERSEDING
MIL-D-51388(EA)
17 July 1974

MILITARY SPECIFICATION

DETECTOR CELL, CHEMICAL AGENT

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of chemical agent detector cell.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-O-20582	-	Opener, Can, Hand, Folding.
MIL-B-117	-	Bag, Sleeve and Tubing, Interior Packaging.
MIL-P-26514	-	Polyurethane Foam, Rigid or Plastic For Packaging.

FEDERAL

PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

FSC 6665

: Beneficial comments (recommendations, add-tions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-123 - Marking for Domestic Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL 5-15-4500 - Detector Cell, Chemical Agent.
E136-91-1520 - Portable Gas Generator, Q219.
D136-40-79-1 - Tester, Agent, Conversion and Detection, Q197A1.
E136-41-2136 - Tester, Leakage and Resistance, Filter and Cell Assemblies, Q198.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 5-15-4500 and on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the detector cell shall conform to the specifications and drawings listed on TDPL 5-15-4500 and subsidiary drawings.

3.2 Manufacture and assembly. The detector cell shall be manufactured and assembled as specified on Drawing D5-15-4500 (see 6.3).

3.3 Blockage. The detector cell shall be checked for blockage of the internal passages from the solution inlet to the air-solution outlet with an airflow rate of 0.20 ± 0.05 liter per minute. There shall be not greater than 10 inches of water pressure drop across the detector cell assembly. The test shall be performed as specified in 4.4.4.1.

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3.4 Air leakage. When tested as specified in 4.4.4.2 the air leakage of the detector cell assembly shall not be greater than 0.30 cubic centimeters per minute with the internal cell pressure increased to 49.0 ± 1.0 inch of water above atmospheric.

3.5 Performance.

3.5.1 Background, noise, and drift. The detector cell shall have a stable electrical background of at least 0.0 millivolt but no greater than 15.0 millivolts during a two-minute monitoring period, and shall not exhibit noise in excess of 2.0 millivolts peak-to-peak during a ten-minute monitoring period or drift in excess of 5.0 millivolts in one direction during a two-hour monitoring period when tested as specified in 4.4.4.3. For the purpose of this paragraph, noise is defined as variations in cell electrical output of 1.0 minute or less in duration, and drift is defined as a longer term change (over 1.0 minute) in the level of the mean electrical output of the cell in either the plus or the minus direction.

3.5.2 Sensitivity. The detector cell shall respond to a concentration of 0.70 ± 0.10 gamma per liter of methane sulfonyl fluoride (MSF), Eastman grade, item 8368 (see 6.4), when tested as specified in 4.4.4.4. The average electrical output of the cell (computed from three exposures to MSF) shall increase no less than 21.0 millivolts above background (see 3.5.1) within 2.0 minutes after start of exposure to MSF.

3.6 Low temperature storage. The detector cell shall meet the requirements of 3.3, 3.4, and 3.5 after the packaged cell has been stored at a temperature of -55° ± 5° F (-48° ± 2.8°C) for a minimum of 14 days as specified in 4.3.3.1. After the storage at low temperature and prior to testing for the requirements of 3.3, 3.4, and 3.5, the packaged detector cell shall be conditioned at 75° ± 10° F (23.9° ± 5.6°C) for at least 14 hours if the detector cell is packaged in accordance with 5.1.1 or for at least 5 hours if the detector cell is packaged in accordance with 5.1.2. When the detector cell is subjected to the temperature parameters specified above, the rate of temperature change shall not exceed 18° F per minute.

3.7 High temperature storage. The detector cell shall meet the requirements of 3.3, 3.4, and 3.5 after the packaged cell has been stored at a temperature of 155° ± 5° F (68.3° ± 2.8°C) for a minimum of 14 days as specified in 4.3.3.2. After the storage at high temperature and prior to testing for the requirements of 3.3, 3.4, and 3.5, the packaged detector cell shall be conditioned at 75° ± 10° F (23.9° ± 5.6°C) for at least 14 hours if the detector cell is packaged in accordance with 5.1.1 or for at least 5 hours if the detector cell is packaged in accordance with 5.1.2. When the detector cell is subjected to the temperature parameters specified above, the rate of temperature change shall not exceed 18° F per minute.

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3.8 Preproduction. Prior to the start of regular production a preproduction sample of chemical agent detector cells shall be produced in accordance with this specification for examination and test (see 4.3).

3.9 Workmanship.

3.9.1 Major defects. The exterior air/solution sealing surfaces of the detector cell shall be free from contamination (chips, dirt, oil, grease, or other foreign materials), damage (cracks or deformation) and adhesive.

3.9.2 Minor defects. The detector cell shall be free from contamination in nonfunctional areas (those which will not cause leakage or separation of parts).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 20 detector cells shall be produced using the same materials, methods, equipment and processes as will be used in regular production including preservation in accordance with Drawing D5-15-4500. The detector cells shall be packaged in accordance with either 5.1.1 or 5.1.2.

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4.3.2 Inspection procedures.

4.3.2.1 For examination. The preproduction sample of detector cells shall be examined in accordance with the classification of defects 4.4.3.3.

4.3.2.2 For test. The preproduction sample of detector cells shall be divided into two equal groups. One group shall be tested in accordance with 4.3.3.1 and the other group shall be tested in accordance with 4.3.3.2.

4.3.3 Tests. Tests shall be conducted as follows:

4.3.3.1 Low temperature storage. Place the packaged detector cells into a temperature test chamber. Both the detector cells and the test chamber shall be at room temperature. Adjust the test chamber to produce the lower temperature required by 3.6. Maintain this temperature for at least 14 days. Adjust the test chamber to produce a temperature of 75° ± 10° F (23.9° ± 5.6°C). Maintain this temperature for at least 5 hours if detector cells were packaged in accordance with 5.1.2 or for at least 14 hours if the detector cells were packaged in accordance with 5.1.1. Remove the detector cells from the test chamber. Subject the detector cells to the blockage, air leakage, and performance tests described in 4.4.4.1, 4.4.4.2, 4.4.4.3, and 4.4.4.4. Blockage, air leakage, and performance shall be as required by 3.3, 3.4, and 3.5.

4.3.3.2 High temperature storage. Place the packaged detector cells into a temperature test chamber. Both the detector cells and the test chamber shall be at room temperature. Adjust the test chamber to produce the higher temperature required by 3.7. Maintain this temperature for at least 14 days. Adjust the test chamber to produce a temperature of 75° ± 10° F (23.9° ± 5.6°C). Maintain this temperature for at least 5 hours if the detector cells were packaged in accordance with 5.1.2 or for at least 14 hours if the detector cells were packaged in accordance with 5.1.1. Remove the detector cells from the test chamber. Subject the detector cells to the blockage, air leakage, and performance tests described in 4.4.4.1, 4.4.4.2, 4.4.4.3, and 4.4.4.4. Blockage, air leakage, and performance shall be as required by 3.3, 3.4, and 3.5.

4.3.4 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

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4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the detector cells offered for acceptance at one time which have been produced by one manufacturer, at one plant, under essentially the same manufacturing conditions and techniques and with no change in materials.

4.4.2 Sampling for examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedures.

4.4.3.1 For examination. Sample detector cells shall be examined in accordance with the classification of defects 4.4.3.3 and MIL-STD-105.

4.4.3.2 For test. Each detector cell in the lot shall be tested in accordance with 4.4.4.1 through 4.4.4.4.

4.4.3.3 Classification of defects.

(a) Detector cell, chemical agent (Drawing D5-15-4500).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing, incorrect, or incorrectly assembled
102	Joint line between the scrubber base and base of scrubber housing, not properly cemented
103	Solution inlet port at base of helical coil blocked
104	Silver metal discolored
105	Large voids in silver spiral
106	RTV at base of terminallug incorrect
107	Joint between cell top and cell center cemented improperly
108	Joint between cell top and cell center and cell scrubber assembly cemented improperly
109	Joint between scrubber spiral and scurbber housing cemented improperly
110	Workmanship (3.9.1)

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4.4.3.3 Classification of defects (continued).

Minor:	AQL 2.5 percent defective
201	Workmanship (see 3.9.2)
202	Marking missing or illegible

(b) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Preservation not as specified
102	Packaging damaged, torn or not as specified
103	Marking missing, incorrect, or illegible

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Blockage. Determine the pressure drop utilizing the Q198 (E136-41-2136) in accordance with the instruction manual.

4.4.4.2 Air leakage. Determine the air leakage utilizing the Q198 (E136-41-2136) in accordance with the instruction manual.

4.4.4.3 Background, noise, and drift. Install the detector cell into a test device which will pump 0.2 ± 0.02 ml of oxime per minute and 1.1 ± 0.2 l of filtered air through the cell. Connect a strip chart recorder with an input impedance of 200,000 ohms or greater across the cell terminals. If the input impedance of the recorder is greater than 200,000 ohms, connect a suitable resistor in parallel with the recorder to give an effective input impedance of 200,000 ohms. After at least 10.0 minutes of operation, re-zero the recorder. Record the output of the detector cell for a period of at least 2.25 hours. The last 2.0 hours of this recording shall meet the drift requirement of 3.5.1, the last 10.0 minutes shall meet the noise requirement of 3.5.1, and the last 2.0 minutes shall meet the background requirement of 3.5.1.

4.4.4.4 Sensitivity. Determine the sensitivity of the cell by utilizing the Q219 (E136-91-1520) and Q197A1 (D136-40-79-1) in accordance with their instruction manuals. Determine the concentration of MSF coming from the dilution apparatus prior to running any sensitivity test, after every four hours during testing and on completion of testing.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, level A. The detector cell shall be preserved and packaged in accordance with paragraph 5.4.

5.2 Packing, interplant shipment (see 6.5). The detector cells, packaged as specified in 5.1, shall be packed to provide physical protection from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall comply with the rules and regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, packages in 5.1, and shipping containers in 5.2, shall be marked in accordance with Fed-Std-123 and shall include the name, address and contract number of the prime contractor.

5.4 Repair parts. When this item is being procured for storage and issue as a repair part, preservation, packing, packaging and marking shall be as specified on the packaging data sheet which is identified by its National stock number.

6. NOTES

6.1 Intended use. The chemical agent detector cell is used as a component of the detector unit ABCA-M43.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Preproduction

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Manufacture of detector cell. Edgewood Arsenal Directive No. 639 is a suggested method of manufacturing detector cells. However, this method is not mandatory for use and in no way relieves the contractor from performing the inspections as set forth herein.

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6.4 Test reagent. Methane sulfonyl fluoride (MSF), Eastman grade, item 8368, boiling point 123° - 124° centigrade, may be purchased from "Distillation Products Industries, Eastman Organic Chemicals Department, Rochester, NY 14603."

6.5 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 251. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

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